

KNOT CONTACT HOMOLOGY AS A PLANAR LIMIT OF CHERN-SIMONS THEORY

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ABSTRACT. We prove a conjecture relating augmentation varieties to the large N limit of Chern-Simons theory. Although this does not directly establish that the augmentation polynomial of a knot is the classical limit of a deformed $\hat{A}$ -polynomial—as suggested by Aganagić and Vafa—it reduces the problem to characterizing certain algebraic properties of a module over the quantum torus, introduced in work of Gaiotto, Kannagi, and Sanjurjo. We term this the *HOMFLYPT difference module*, which captures relations between the colored HOMFLYPT polynomials of different antisymmetric colorings. We demonstrate that the classical limit of this difference module for a knot is precisely the degree 0 abelianized knot contact homology of the knot, and we provide a natural extension of this result to links.

1. INTRODUCTION

In [GKS25], Gaiotto, Kannagi, and Sanjurjo discuss applications of holographic duality between Chern-Simons theory and the A-model of topological string theory. The former has well-known applications in knot theory as the source of quantum invariants, such as the Jones polynomial; on the dual side, this should manifest as the study of open Gromov–Witten theory for the conormal to the knot in T^*S^3 . This, in turn, has been related to knot contact homology by the work of Aganagić, Ekholm, Ng, and Vafa [Vaf+14]. From a mathematical perspective, this leads to the conjecture that the three-variable augmentation variety of a link can be described in terms of a large N limit of Chern-Simons theory for $\mathfrak{sl}_N$.

More precisely, by the work of Garoufalidis, Lauda, and Lê [GLL18], the HOMFLYPT polynomials of a link corresponding to the wedge powers $\bigwedge^m \mathbb{C}^N$ form a q -holonomic sequence in m . That is, the sequence satisfies a recursion relation in the q -commuting transformations Λ , which sends $m \mapsto m - 1$, and ν , which acts by multiplication by q^m . This relation is sometimes called the deformed $\hat{A}$ -polynomial.

Based on [GKS25], we reinterpret this as a statement about a **HOMFLYPT difference module**: a module over the quantum torus that provides a purely skein-theoretic description of these recursions by treating m as a formal variable. While interpreting the polynomial as a function of q for formal m presents difficulties, one can work skein-theoretically to relate values where m is shifted by an integer.

We illustrate this with an example based on [GKS25, (2.15) & §2.3]. Consider the unknot labeled with the representation $\bigwedge^m \mathbb{C}^N$. We can compare this with the value of the quantum knot invariant for $\bigwedge^{m-1} \mathbb{C}^N$ using relations between webs in MOY (Murakami–Ohtsuki–Yamada) calculus, a diagrammatic calculus for quantum $\mathfrak{gl}_N$ invariants that we review in detail in Section 2.1:

$$\text{Unknot with label } m = \frac{q - q^{-1}}{q^m - q^{-m}} \text{ (crossing) } m = \frac{q^{N-m+1} - q^{-N+m-1}}{q^m - q^{-m}} \text{ (cup) } m-1.$$

We can “HOMFLYPTize” this formula in the final step by substituting $g = q^N$ and $\nu = q^m$. This substitution is purely formal: while evaluating the HOMFLYPT polynomial at a non-integer value of m is problematic, the skein relations themselves remain valid when we treat $\nu = q^m$ as an independent variable. The key insight is that the MOY relations relate diagrams with labels differing by integers, so we can work with formal shifts of m without ever needing to evaluate at a specific value. This removes the explicit dependence on N , and now the equation only depends on m through ν . This is effectively taking the planar limit of Chern-Simons theory, sending $N \rightarrow \infty$ and $q \rightarrow 1$.

Finally, since m is now purely formal, we can define a transformation Λ (the precise domain and codomain of which are discussed later) that sends the label m to $m - 1$. While this shift is not well-defined for fixed integers m , it is consistent for a formal variable. To preserve the MOY relations—specifically (2.3)—the transformations Λ and ν must q -commute such that $\Lambda\nu = q\nu\Lambda$. Using this

notation, we have:

$$\bigcirc^m = \frac{qg\nu^{-1} - q^{-1}g^{-1}\nu}{\nu - \nu^{-1}} \bigcirc^{m-1} = \frac{qg\nu^{-1} - q^{-1}g^{-1}\nu}{\nu - \nu^{-1}} \Lambda \bigcirc^m$$

Thus, we conclude that the expression $\nu - \nu^{-1} - \Lambda(g\nu^{-1} - g^{-1}\nu)$ annihilates the value of the unknot. This is the deformed $\hat{A}$ -polynomial of the unknot, up to several changes of convention. For instance, our results align with [AV12, §6.1], with minor differences due to their use of symmetric rather than antisymmetric powers; our variables q, g, ν, Λ correspond to their $q^{1/2}, Q^{1/2}, e^{\hat{p}/2}, e^{\hat{x}}$, respectively. After setting $q = 1$ and applying the substitution in (3.1), this matches the augmentation polynomial of the unknot. Note that this derivation does not require an explicit calculation of the quantum invariant; in Remark 2.7, we discuss the challenges of defining that invariant in the limits $N \rightarrow \infty$ and $q \rightarrow 1$.

Fix a general link L . Consider the quantum torus

$$\mathcal{T}_r = \mathbb{C}[q^{\pm 1}, g^{\pm 1}] \langle \Lambda_i, \nu_i \rangle / (\Lambda_i \nu_i = q^{\delta_{ij}} \nu_i \Lambda_j, \nu_i \nu_j = \nu_j \nu_i, \Lambda_i \Lambda_j = \Lambda_j \Lambda_i)$$

whose number of variables r is equal to the number of components of L ; let $\mathcal{N}_r$ be the subalgebra generated by $q^{\pm 1}$ and ν_i . Applying an argument analogous to that above, we obtain a module $H_q(L)$ over $\mathcal{T}_r$ whose elements are webs where the knot is colored by formal parameters m_i shifted by integers, with additional web portions colored by integers. We call this the **HOMFLYPT difference module** of L .

Let $\mathcal{F}_r$ be the space of all maps from $\mathbb{Z}^r \rightarrow \mathbb{C}(q)[g^{\pm 1}]$. This is a module over the quantum torus $\mathcal{T}_r$ by the action

$$\nu_i f(k_1, \dots, k_r) = q^{k_i} f(k_1, \dots, k_r), \quad \Lambda_i f(k_1, \dots, k_r) = f(k_1, \dots, k_i - 1, \dots, k_r).$$

There is a natural $\mathcal{T}_r$ -module homomorphism $\text{ev}: H_q(L) \rightarrow \mathcal{F}_r$ sending a web w to the function $f_w(k_1, \dots, k_r)$ given by the evaluation of w in the HOMFLYPT skein category after substituting k_i for m_i . This provides a new perspective on finding recursions satisfied by antisymmetric HOMFLYPT polynomials by framing the problem as an investigation into the structure of the HOMFLYPT difference module $H_q(L)$. Let $[L]$ be the element of $H_q(L)$ consisting of the link L colored by the parameters m_i .

Definition 1.1. Let $I_q(L) = \{x \in \mathcal{T}_r \mid x[L] = 0\}$ be the annihilator of $[L]$.

Since the homomorphism ev maps $[L]$ to the colored HOMFLYPT polynomials of L for all antisymmetric colorings (viewed as an element of $\mathcal{F}_r$), all elements of the left ideal $I_q(L)$ yield recurrence relations satisfied by these polynomials. We conjecture (Conjecture 2.17) that ev is injective and that $I_q(L)$ coincides with the ideal $I_q(L)'$ of recurrence relations satisfied by the antisymmetric HOMFLYPT polynomials of L .

The q -holonomicity theorem of [GLL18] shows that the variety defined by $I_q(L)'$ in the classical limit $q \rightarrow 1$ is Lagrangian; accordingly, one would expect the same for $I_q(L)$, suggesting that $H_q(L)$ is “not too big.”

The advantage of this perspective is that, unlike its image under ev , we can provide an explicit presentation of the module $H_q(L)$.

We can write $H_q(L)$ as a quotient of the tensor product of the quantum torus and enveloping algebra $U = \mathcal{T}_r \otimes U_q(\mathfrak{gl}_s)$ arising from a presentation of β as the closure of a braid with s strands. Specifically, it is a quotient of this algebra by the sum of a left and a right ideal—that is, a tensor product of cyclic left and right modules. The resulting presentation has a particularly simple classical limit as $q \rightarrow 1$, consistent with earlier discussions of the augmentation polynomial.

Theorem A. The base change $H(L)$ of $H_q(L)$ to the locus where $q = 1$ (keeping ν_i, Λ_i as formal variables) is the degree 0 part of the abelianized knot contact homology $\text{KCH}(L)$ under the parameter change:

$$(1.1) \quad \mu \mapsto \nu^{-2}, \quad U \mapsto g^{-2}, \quad \lambda \mapsto -g^{-1} \Lambda^{-1}.$$

In particular, the base change $I(L)$ of the ideal $I_q(L)$ lies in the augmentation ideal $\text{Au}_L \subset \mathbb{C}[U^{\pm 1}, \mu_i^{\pm 1}, \lambda_i^{\pm 1}]$ of L .

We conjecture that ev is injective, so $I_q(L)$ describes the recursions of the colored HOMFLYPT polynomials and that this base change is precisely the augmentation ideal of L (see Conjecture 3.12).

Together, these conjectures would provide a rigorous proof of the conjecture in [AV12; Vaf+14] that the augmentation ideal of L is a classical limit of the recursions satisfied by colored HOMFLYPT polynomials.

We emphasize that the proof of Theorem A is purely algebraic. Compared to the difficulty of calculating knot invariants for specific values of N , our approach is relatively elementary and follows the spirit of [GLL18]: one derives recursions for quantum knot invariants from MOY calculus and demonstrates, without explicit calculation of invariants, that in the limit $N \rightarrow \infty, q \rightarrow 1$, these relations recover the equations of the augmentation variety implicitly given in [Ekh+13, Th. 1.3].

Formalizing these arguments requires working with a limit of the representation category of $U_q(\mathfrak{gl}_N)$ where, in addition to standard wedge powers $\bigwedge^m \mathbb{C}^N$, we include objects where m is treated as a formal variable. This limit appears worthy of further study, although it breaks certain structures of the category, such as its braided and rigid structures.

An index of notation is provided at the end of the paper for reference.

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2. HEAVY SKEIN ALGEBRAS

2.1. The HOMFLYPT spider. Our starting point is the ribbon category of representations of $U_q(\mathfrak{gl}_N)$. This is a braided spherical $\mathbb{C}[q, q^{-1}]$ -linear category.

However, in this paper, rather than considering N as a fixed integer, we vary it, following the approach of Deligne categories. The key idea, going back to Deligne’s work on interpolating representation categories [deligneCategorieTannakienne2007], is that many categorical constructions that appear to require N to be a positive integer can be defined for formal values of N by working with polynomial relations rather than explicit vector spaces. A version of the Deligne category for $U_q(\mathfrak{gl}_N)$ is discussed by Brundan in [Bru17, §1.4], based on the oriented skein category. We will use a variant of this approach based on the work of Murakami, Ohtsuki, and Yamada [MOY98], developed further by Cautis, Kamnitzer, and Morrison [CKM14] and Queffelec and Sartori [QS19], which gives a combinatorial description of this category in terms of *spiders*, certain planar diagrams on which we impose linear relations. This makes it clear how the relations of the category depend on N . In fact, we can remove the dependence on N entirely by considering the power $g = q^N$ as an independent variable—one can think of this as “HOMFLYPTizing” the spider category, since the resulting knot invariants become functions of q and g independently, as in the HOMFLYPT polynomial. The resulting category is denoted $\mathrm{Sp}(N)$ in the notation of [QS19, Def. 6.5].

The variable N only appears in the spider relations in quantum binomial coefficients of the form

$$(2.1) \quad \begin{bmatrix} N-a \\ b \end{bmatrix}_q = \frac{(gq^{-a} - g^{-1}q^a) \cdots (gq^{-a-b+1} - g^{-1}q^{a+b-1})}{(q^b - q^{-b}) \cdots (q - q^{-1})}.$$

We will continue to use the symbol $\begin{bmatrix} N-a \\ b \end{bmatrix}_q$, but use it to mean the rational function in g and q given by (2.1).

The category $\mathrm{Sp}(N)$ is defined in terms of ladders:

Definition 2.1. *A ladder with k uprights is a diagram drawn in a rectangle with*

- *k parallel vertical lines that run from the bottom edge to the top edge of the rectangle, oriented upwards or downwards.*
- *some number of oriented horizontal rungs connecting adjacent uprights,*
- *a labeling of each rung by a non-negative integer and of each section of an upright by an element of $\mathbb{Z}_{\geq 0}$. Labeling must satisfy the condition that the signed sum of the segments at each vertex is zero.*

Definition 2.2. *The (HOMFLYPT) spider category $\mathrm{Sp}(N)$ is a braided monoidal category generated by objects $\mathbb{Z}_{\geq 0}^+ = \{n^+ \mid n \in \mathbb{Z}_{\geq 0}\}$ and their duals $\mathbb{Z}_{\geq 0}^- = \{n^- \mid n \in \mathbb{Z}_{\geq 0}\}$. In the original*

representation-theoretic interpretation of this category, the objects $n^\pm$ corresponded to $\bigwedge^n V$ and its dual $(\bigwedge^n V)^*$.

Thus, a general object in this category is a finite list of these generators, which one should think of as corresponding to a tensor product of the wedge powers mentioned above.

The tensor product of two objects is given by concatenating their lists; in particular, the empty list gives a unit object in the category. We include an isomorphism of $0^\pm$ to the empty list in our category.

The morphisms from $(a_1^{\epsilon_1}, \dots, a_m^{\epsilon_m})$ to $(b_1^{\tau_1}, \dots, b_k^{\tau_k})$ for $a_i, b_i \in \mathbb{Z}_{>0}$ and $\epsilon_i, \tau_i \in \{+, -\}$ are generated by a fixed isomorphism $0^+ \cong \emptyset$ and by ladders in $\mathbb{R} \times [0, 1]$ that satisfy:

- (1) The points where the graph meets $\mathbb{R} \times \{0\}$ are labeled with a_i read from left to right, with the edge oriented upward if $\epsilon_i = +$ and downward if $\epsilon_i = -$.
- (2) The points where the graph meets $\mathbb{R} \times \{1\}$ are labeled with b_i read from left to right, with the edge oriented upward if $\tau_i = +$ and downward if $\tau_i = -$.

The tensor product of morphisms is as usual given by horizontal composition. We can interpret an arbitrary morphism in this category as a trivalent graph, where applying the isomorphism $0^+ \cong \emptyset$ to an upright of a ladder is represented by simply contracting that upright and smoothing out the trivalent vertex it arose from. Conversely, we can “ladderize” any such trivalent graph to obtain a unique morphism, using uprights with the label $0^\pm$.

The morphism space between two objects is spanned by formal linear combinations of such graphs over $\mathbb{C}(q)[g^{\pm 1}]$, considered up to isotopy of the corresponding graphs, modulo the relations below and their mirror images and rotations.

$$(2.2) \quad \text{circle with arrow} = \begin{bmatrix} N \\ k \end{bmatrix}_q$$

$$(2.3) \quad \text{graph with loop} = \begin{bmatrix} k \\ l \end{bmatrix}_q \text{ graph} \quad \text{graph with loop} = \begin{bmatrix} N-k \\ l \end{bmatrix}_q \text{ graph}$$

$$(2.4) \quad \text{trivalent graph} = \text{trivalent graph}$$

$$(2.5) \quad \text{graph} = \sum_t \begin{bmatrix} k-l+r-s \\ t \end{bmatrix}_q \text{ graph}$$

$$(2.6) \quad \text{graph} = \sum_t \begin{bmatrix} N-k-l+r-s \\ t \end{bmatrix}_q \text{ graph}$$

$$(2.7) \quad \begin{array}{c} k-s-r \quad l+s+r \\ \uparrow \quad \uparrow \\ \text{Diagram 1} \\ \uparrow \quad \uparrow \\ k-s \quad l+s \\ \uparrow \quad \uparrow \\ k \quad l \end{array} = \begin{bmatrix} r+s \\ r \end{bmatrix}_q \begin{array}{c} k-s-r \quad l+s+r \\ \uparrow \quad \uparrow \\ \text{Diagram 2} \\ \uparrow \quad \uparrow \\ k \quad l \end{array}$$

$$(2.8) \quad \begin{array}{c} k_2+r+s \quad k_3-s \\ \uparrow \quad \uparrow \\ \text{Diagram 3} \\ \uparrow \quad \uparrow \\ k_1-r \quad k_2+r \\ \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} = \begin{array}{c} k_1-r \quad k_2+r+s \\ \uparrow \quad \uparrow \\ \text{Diagram 4} \\ \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array}$$

$$(2.9) \quad \begin{array}{c} k_2-r-s \quad k_3+s \\ \uparrow \quad \uparrow \\ \text{Diagram 5} \\ \uparrow \quad \uparrow \\ k_1+r \quad k_2-r \\ \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} = \begin{array}{c} k_1+r \quad k_2-r-s \\ \uparrow \quad \uparrow \\ \text{Diagram 6} \\ \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array}$$

$$(2.10) \quad \begin{array}{c} k_1-2 \quad k_2+1 \quad k_3+1 \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{Diagram 7} \\ \uparrow \quad \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} - [2]_q \begin{array}{c} k_1-2 \quad k_2+1 \quad k_3+1 \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{Diagram 8} \\ \uparrow \quad \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} + \begin{array}{c} k_1-2 \quad k_2+1 \quad k_3+1 \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{Diagram 9} \\ \uparrow \quad \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} = 0$$

$$(2.11) \quad \begin{array}{c} k_1-1 \quad k_2-1 \quad k_3+2 \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{Diagram 10} \\ \uparrow \quad \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} - [2]_q \begin{array}{c} k_1-1 \quad k_2-1 \quad k_3+2 \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{Diagram 11} \\ \uparrow \quad \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} + \begin{array}{c} k_1-1 \quad k_2-1 \quad k_3+2 \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{Diagram 12} \\ \uparrow \quad \uparrow \quad \uparrow \\ k_1 \quad k_2 \quad k_3 \end{array} = 0$$

We can define over and undercrossings in this category by the formulas below, making it into a braided monoidal category:

$$(2.12) \quad \begin{array}{c} l \quad k \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ k \quad l \end{array} = \sum_{\substack{a, b \geq 0 \\ b-a=k-l}} (-q)^{k-b} k-b \quad \begin{array}{c} l \quad k \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ k \quad l \end{array} = \sum_{\substack{a, b \geq 0 \\ b-a=l-k}} (-q)^{l-b} k+b$$

$$(2.13) \quad \begin{array}{c} l \quad k \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ k \quad l \end{array} = \sum_{\substack{a, b \geq 0 \\ b-a=k-l}} (-q)^{b-k} k-b \quad \begin{array}{c} l \quad k \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ k \quad l \end{array} = \sum_{\substack{a, b \geq 0 \\ b-a=l-k}} (-q)^{b-l} k+b$$

The spider category is closely related to another diagrammatic category:

Lemma 2.3. *There is an inclusion functor $\mathcal{OS}(q^{-1} - q, -g) \rightarrow \mathbf{Sp}(N)$ and this functor induces an equivalence of Karoubi envelopes.*

Here $\mathcal{OS}(z, t)$ denotes the **oriented skein category**, whose morphisms are tangles in a thickened strip modulo the oriented skein relation; see [Bru17, §3] for details. This lemma, noted in [Bru17] and proved in [QS19, Prop. 6.7], shows that our spider category is equivalent (after idempotent completion) to the category underlying the HOMFLYPT skein theory.

In particular, we have a skein relation generalizing [Bru17, (S)]:

$$(2.14) \quad \begin{array}{c} \ell \quad 1 \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ 1 \quad \ell \end{array} - \begin{array}{c} \ell \quad 1 \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ 1 \quad \ell \end{array} = (q^{-1} - q) \begin{array}{c} \ell \quad 1 \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ 1 \quad \ell \end{array}$$

As mentioned before, these relations give an isomorphism between $0^\pm$ and the empty list via the lollipop diagram (a single trivalent vertex, with two of the adjacent edges joined). Thus, our category is effectively unchanged if we just delete all edges with label 0.

It follows by [QS19, Cor. 6.17] and [Bru17, Th. 1.3] that we can recover the usual finite $\mathfrak{gl}_N$ spider category $\mathbf{Sp}(e)_{\leq e}$ for $e \in \mathbb{Z}_{\geq 0}$ by specializing $g = q^e$ and killing the objects $n^\pm$ for all $n > e$. Note that this specialization already sets the dimension of the objects we have killed to 0.

Experts will note that we have used a renormalized braiding from the usual choice for $\mathfrak{sl}_N$ and thus (2.12) does not match [GKS25, (E.6)]; this does match the renormalized crossing introduced in [GKS25, (C.8)]. This results in the same knot invariants for knots in the zero-framing as the usual choice, but at the price that we can only choose a realization functor to the braided category of $U_q(\mathfrak{gl}_N)$ modules; as noted by Queffelec [Que15], we need the strand with label N to correspond to a non-trivial object. This difference is not important for our purposes, and it will simplify things for us to avoid these powers.

An important observation that helps us to structure this category is [CKM14, Prop. 5.2.1]: there is a functor from the idempotent form $\tilde{U}$ of $U_q(\mathfrak{gl}_k)$, considered as a category with each weight as an object. This functor sends the weight $(p_1, \dots, p_k)$ to the tensor product $p_1^+ \otimes \dots \otimes p_k^+$ if $p_i \geq 0$ for all i , and 0 otherwise. Composing with the realization functor, we recover skew Howe duality ([CKM14, Th. 4.2.1]). We can generalize this to include the objects n^- as in [QS19] (where they denote n^+ by n and n^- by n^*); in this case, we still have a functor from a version of the idempotent form of $U_q(\mathfrak{gl}_N)$, but the relations Eqs. (2.3) and (2.6) show that we must send n^- to $N - n$ in the weight of the quantum group as in [QS19, (7.4)].

2.2. Heavy objects. This subsection introduces heavy uprights and explains their role in the planar limit, giving a brief intuition for readers less familiar with Deligne-style limits.

We wish to expand this category by adding new objects which are “heavy” to contrast with the existing objects which are “light.” This means that rather than an integer, they correspond to a formal label m in an arbitrary index set M , or more generally, this formal label plus an integer (of either sign). As befits their name, these objects will be represented by a thicker line than the light ones.

Before discussing formalities, let us indicate the intuition behind this construction. Our work is based on [GKS25, §2.2], where these heavy objects are considered in the context of the N

to

infty limit of Chern-Simons theory. The key distinction is:

beginitemize

item

textbfLight objects correspond to representations

$bigwedge^{1k}$

C^N where k is a fixed integer independent of N . The quantum dimension of such an object is a polynomial in $g = q^N$.

item

textbfHeavy objects correspond to representations

$bigwedge^{1\alpha N}$

C^N where

α

in $(0, 1)$ is fixed, so the label scales with N . In the planar limit, these become the “D-branes” of the dual string theory.

enditemize For our purposes, it is better to think about

αN as an independent formal variable m , but in future work, we hope to do a more detailed analysis in view of the scaling interpretation.

We work over the base ring $\mathbb{k} = \mathbb{C}(q)[g^{\pm 1}, q^{\pm m}]_{m \in M}$. When we have a list of elements $m_1, \dots, m_k \in M$, we write $\nu_i = q^{m_i}$ for simplicity of notation. In [GKS25], the symbol μ_i is used for ν_i , but we wish to avoid confusion with the notation for knot contact homology, since a change of variables is needed to relate to the usual notation, for example in [Ng14]. As before, we use N as just a formal symbol that reminds us to add factors of $g = q^N$.

We consider the ladders as in Definition 2.1, but now using labels from the set M :

Definition 2.4. A **heavy ladder** with k uprights is a ladder diagram where each rung by a non-negative integer and of each section of an upright is labeled by an element of $\mathbb{Z}_{\geq 0}$, or an element of $M + \mathbb{Z}$. The labeling must satisfy the condition that the signed sum of the segments at each vertex is zero. Note that this means that all the labels on a single upright lie in $m + \mathbb{Z} + kN$ for a fixed $m \in M$ or in $\mathbb{Z}_{\geq 0}$.

Let $\Pi = \mathbb{Z}_{\geq 0} \cup M + \mathbb{Z}$. We call the uprights with labels in $M + \mathbb{Z}$ **heavy** and those with labels in $\mathbb{Z}_{\geq 0}$ **light**.

As we did with the integers, we consider positive and negative copies of Π as labels on the bottom and top of each upright of a ladder: if r is the label on the adjacent portion of the upright, then we label the bottom with r^+ if it is oriented up and r^- if it is oriented down. For convenience, given $r \in \Pi, a \in \mathbb{Z}$, we let $r^\pm + a = (r \pm a)^\pm$.

Definition 2.5. As in the definition of $\mathrm{Sp}(N)$, we consider the monoidal category $\mathrm{Sp}_M(N)$ generated by objects Π^+ and Π^- , so that a general object is a list of these generators, with tensor product given by concatenation.

The morphisms from $(a_1^{\epsilon_1}, \dots, a_m^{\epsilon_m})$ to $(b_1^{\tau_1}, \dots, b_k^{\tau_k})$ for $a_i, b_i \in \Pi$ and $\epsilon_i, \tau_i \in \{+, -\}$ are generated by a fixed isomorphism $0^+ \cong \emptyset$ and by heavy ladders in $\mathbb{R} \times [0, 1]$ that satisfy:

- (1) The points where the ladder meets $\mathbb{R} \times \{0\}$ are labeled with a_i read from left to right, with the edge oriented upwards if $\epsilon_i = +$ and downwards if $\epsilon_i = -$.
- (2) The points where the ladder meets $\mathbb{R} \times \{1\}$ are labeled with b_i read from left to right, with the edge oriented upward if $\tau_i = +$ and downward if $\tau_i = -$.

We can convert these ladders into graphs as in Definition 2.2. The morphism space from the object $(a_1^{\epsilon_1}, \dots, a_m^{\epsilon_m})$ to $(b_1^{\tau_1}, \dots, b_k^{\tau_k})$ is thus spanned by the graphs arising from heavy ladders as above, modulo the local relations (2.2–2.11) with any mix of heavy and light uprights, where we interpret the q -binomial

coefficient as a formal rational function in terms of $q, g = q^N, \nu = q^m$ and $\nu' = q^{m'}$ as in (2.1):

$$(2.15) \quad \left[\begin{matrix} m - m' + r - s \\ t \end{matrix} \right]_q = \frac{\prod_{i=0}^{t-1} q^{r-s-i} \nu(\nu')^{-1} - q^{s-r+i} \nu^{-1} \nu'}{\prod_{i=1}^t q^i - q^{-i}}$$

$$(2.16) \quad \left[\begin{matrix} N - m - m' + r - s \\ t \end{matrix} \right]_q = \frac{\prod_{i=0}^{t-1} g q^{r-s-i} (\nu \nu')^{-1} - g^{-1} q^{s-r+i} \nu \nu'}{\prod_{i=1}^t q^i - q^{-i}}$$

We let $\mathbf{m}, \epsilon$ be the k -tuples in M (ignoring addition by integers) and $\{\pm\}$ such that $m_i^{\epsilon_i}$ is the label on the i th heavy upright. Since we do not allow heavy labels on rungs, if these sequences are different, then the morphism space between the corresponding objects is trivial.

Definition 2.6. Let $\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$ be the full subcategory of $\mathbf{Sp}_M(N)$ consisting of the objects $((m_1 + p_1)^{\epsilon_1}, \dots, (m_n + p_n)^{\epsilon_n})$ with $p_i \in \mathbb{Z}$. The morphisms in this category can be written as ladders where all uprights are heavy.

Let $\mathbf{Spl}_{\mathbf{m}, \epsilon}(N)$ be the larger category where we allow additional light uprights at the top and bottom of the diagram while leaving the same sequence of heavy uprights. Note that a generic slice with y -value fixed is a morphism in $\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$

Another consequence is that we cannot allow heavy objects to braid with each other or to form cups and caps, so we no longer have a braided or rigid category once we add them. It is an intriguing question in what context we can define such a braiding, which would seem to require an infinite sum.

Remark 2.7. To see some of the difficulties of defining a value for heavy objects, consider the case of the unknot. Note that the argument in the introduction where we found a recursion for the quantum invariants of the unknot did not require us to assign an actual value to the unknot in the limit $N \rightarrow \infty$ and $q \rightarrow 1$; obviously, this is not possible as a single complex number, but there is an interesting way to think about it as a divergent series, discussed in [GKS25, §2.2]. For finite values of N, m , if we fix $\hbar = \frac{\log q}{N}$ and so $q = e^{\hbar}$, one obtains

$$(2.17) \quad \log \bigcirc^m = \sum_{i=0}^{m-1} \log(g q^{-i} - g^{-1} q^i) - \log(q^{i+1} - q^{-i-1})$$

$$(2.18) \quad = \sum_{i=0}^{m-1} \log(g^{1-i/N} - g^{-1+i/N}) - \log(g^{(i+1)/N} - g^{(-i-1)/N})$$

For large m and N , this becomes a Riemann sum approximating the integral

$$N \int_0^{m/N} \log(g^{1-x} - g^{-1+x}) - \log(g^x - g^{-x}) dx$$

We can now treat m and N as continuous quantities, fix the ratio $\alpha = m/N$, and write this as

$$\log \bigcirc^m = \frac{\log g}{\hbar} \int_0^\alpha \log(g^{1-x} - g^{-1+x}) - \log(g^x - g^{-x}) dx$$

Thus, considered as a holomorphic function of $\hbar$, the value of the unknot has an essential singularity at $\hbar = 0$ proportional to the function $e^{1/\hbar}$, but away from this point, it is holomorphic.

The study of how to make sense of these sort of functions is the most essential part of resurgent analysis, so this argument suggests a connection with the study of resurgence in the context of knot invariants [GGM21].

Remark 2.8. One potential approach to making sense of these heavy objects is to consider an ultraproduct category. That is, for every $\mathbf{k} \in \mathbb{Z}^M$, we consider a copy of the category $\mathbf{Sp}(N)$, and consider the ultraproduct of these categories with respect to a Fubini non-principal ultrafilter on $\mathbb{Z}^M$. We can define light objects in this ultraproduct category by taking the same object (a) in each factor of the product for all $\mathbf{k}$ and define a heavy object with label m_i by taking the usual object (k_i) in each factor. The resulting category is linear over the ultraproduct of copies of $\mathbb{C}[q, q^{-1}]$, and still braided and ribbon. The price one pays for this is that the field of scalars is enormous; a closed heavy knot has a well-defined value, but it is just the ultraproduct of the values of the HOMFLYPT polynomial of the knot with different colors.

It is an interesting question to ask whether this ultraproduct category has a natural MOY type model; the relations in $\mathbf{Sp}_{\mathbf{m},\epsilon}(N)$ hold of course, but this does not account for skein type relations involving heavy tangles. In particular, since the tensor product $\bigwedge^a \mathbb{C}^N \otimes \bigwedge^b \mathbb{C}^N$ is multiplicity-free for all a, b, N , we have relations that arise from mutation of knots. This shows that two tangles that differ by a mutation give the same morphism in the ultra-product category by [MC96, Th. 5].

On the other hand, we can make sense of the braiding of heavy and light objects by extending the equations (2.12–2.13) when one of the strands in the formula is light. Let us just record the result in the case where the light strand has the label 1.

$$(2.19) \quad \begin{array}{c} m \quad 1 \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ 1 \quad m \end{array} = \begin{array}{c} m \quad 1 \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ 1 \quad m \end{array} - q^{-1} \begin{array}{c} m \quad 1 \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ 1 \quad m \end{array}$$

$$(2.20) \quad \begin{array}{c} 1 \quad m \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ m \quad 1 \end{array} = \begin{array}{c} 1 \quad m \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ m \quad 1 \end{array} - q \begin{array}{c} 1 \quad m \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ m \quad 1 \end{array}$$

$$(2.21) \quad \begin{array}{c} 1 \quad m \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ m \quad 1 \end{array} = \begin{array}{c} 1 \quad m \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ m \quad 1 \end{array} - q^{-1} \begin{array}{c} 1 \quad m \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ m \quad 1 \end{array}$$

$$(2.22) \quad \begin{array}{c} m \quad 1 \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ 1 \quad m \end{array} = \begin{array}{c} m \quad 1 \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ 1 \quad m \end{array} - q \begin{array}{c} m \quad 1 \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ 1 \quad m \end{array}$$

In particular, we have a thick version of the skein relation (2.14):

$$(2.23) \quad \begin{array}{c} 1 \quad m \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ m \quad 1 \end{array} - \begin{array}{c} 1 \quad m \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ m \quad 1 \end{array} = (q^{-1} - q) \begin{array}{c} 1 \quad m \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ m \quad 1 \end{array}$$

$$(2.24) \quad \begin{array}{c} m \quad 1 \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ 1 \quad m \end{array} - \begin{array}{c} m \quad 1 \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ 1 \quad m \end{array} = (q^{-1} - q) \begin{array}{c} m \quad 1 \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ 1 \quad m \end{array}$$

This makes $\mathbf{Sp}_M(N)$ into a monoidal bimodule category over $\mathbf{Sp}(N)$ with a braiding relating the left and right actions. Since the functors of horizontal composition on the left or right by a given object are isomorphic, we can package this information into just the left action of $\mathbf{Sp}(N)$, which is a braided monoidal action in the sense of [BBJ18, Def. 3.4], with the automorphism E given by winding the object of $\mathbf{Sp}(N)$ around the thick strand in the positive direction. As discussed in [BBJ18, Rmk. 3.6], this notion depends on a framing of the annulus, which gives us the framing of the thin strands as they wrap around the thick one. However, the ribbon structure of $\mathbf{Sp}(N)$ allows us to relate all these different framings by adding in an appropriate number of full twists to the object in $\mathbf{Sp}(N)$ as it wraps around.

Returning to thinking of left and right actions, the braiding structure of the action exactly captures the fact that there are two isomorphisms between the left and right actions, corresponding to the two different ways of braiding the heavy and light objects.

Remark 2.9. From a ribbon category like $\mathbf{Sp}(N)$, we can produce an invariant of surfaces called the *skein category*. This is a natural extension of the skein algebras; it seems to have first appeared in written form in unpublished work of Walker. It was shown by Cooke in [Coo23] that we can interpret this category as the factorization homology of the ribbon category on the surface.

We should be able to interpret our “heavy strands” as taking the factorization homology with marked points as discussed in [BBJ18, §5.1], considering the case with a single heavy upright as a braided module category over $\mathbf{Sp}(N)$, which we then put on each marked point.

The only case that will be important for us is when the surface is $\mathbb{R}^2$ or S^2 . In the former case, we obtain the subcategory of $\mathbf{Sp}_M(N)$ where we fix k heavy uprights with labels in $m_i^{\epsilon_i} + \mathbb{Z}$ for fixed $m_i \in M, \epsilon_i \in \{+, -\}$ and then add in arbitrary light uprights; in the latter case, we obtain a quotient of this ring where we quotient by specializing the central elements of the universal enveloping algebra to the scalars given by the “charge at ∞ ” in [GKS25, §3.1].

It is straightforward to give a basis of the morphism spaces in this category, based on the PBW theorem. First, recall that we can give the quantum group $U_q(\mathfrak{gl}_n; \mathbb{k})$ a grading by the group $\mathbb{Z}^n$, where the homogeneous component for $(w_1, \dots, w_n) \in \mathbb{Z}^n$ is the set of elements of this weight under the adjoint action of the Cartan—let

$$U_{(w_1, \dots, w_n)} = \{u \in U_q(\mathfrak{gl}_n; \mathbb{k}) \mid K_i u K_i^{-1} = q^{w_i} u \text{ for } i = 1, \dots, n\}.$$

Let $q^{r^+} = q^r$ and $q^{r^-} = gq^{-r}$. For a fixed n -tuple $(r_1, \dots, r_n)$ with $r_i \in \Pi^\pm$, consider the map $\mathbb{k}[\mathbf{K}^{\pm 1}] = \mathbb{k}[K_1^{\pm 1}, \dots, K_n^{\pm 1}] \rightarrow \mathbb{k}$ sending $K_i \mapsto q^{r_i}$; let $\mathbb{k}_{\mathbf{r}}$ be $\mathbb{k}$, thought of as a $\mathbb{k}[\mathbf{K}^{\pm 1}]$ -module.

Definition 2.10. Let $\dot{U}_{\mathbf{m}, \epsilon} := \dot{U}_{\mathbf{m}, \epsilon}(\mathfrak{gl}_n; \mathbb{k})$ be the category whose objects are n -tuples $\mathbf{r} = (r_1, \dots, r_n)$ with $r_i \in m_i^{\epsilon_i} + \mathbb{Z}$, and where

$$\mathrm{Hom}(\mathbf{r}, \mathbf{r}') = \mathbb{k}_{\mathbf{r}'} \otimes_{\mathbb{k}[\mathbf{K}^{\pm 1}]} U_q(\mathfrak{gl}_n; \mathbb{k}) \otimes_{\mathbb{k}[\mathbf{K}^{\pm 1}]} \mathbb{k}_{\mathbf{r}} \cong U_{\mathbf{r}' - \mathbf{r}} \otimes_{\mathbb{k}[\mathbf{K}^{\pm 1}]} \mathbb{k}_{\mathbf{r}}$$

By a natural extension of [QS19, Prop. 7.2 & 7.5], we see that we have:

Lemma 2.11. *There is an equivalence of categories*

$$\dot{U}_{\mathbf{m}, \epsilon} \rightarrow \mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$$

sending $\mathbf{r}$ to the same n -tuple, thought of as an object in $\mathbf{Sp}_M(N)$, and sending E_i and F_i to the usual diagrams

$$E_i \mapsto \begin{array}{c} | \quad | \\ \diagdown \quad \diagup \\ 1 \end{array} \quad F_i \mapsto \begin{array}{c} | \quad | \\ \diagup \quad \diagdown \\ 1 \end{array}$$

Proof. The confirmation of the quantum group relations is exactly as in [QS19, Prop. 7.2], showing that we have a functor. This functor is full since any ladder can be written in terms of label 1 rungs; rungs with other integer labels are the images of divided powers $E_i^{(r)}, F_i^{(r)}$. It is also essentially surjective by direct construction. Thus, the only nontrivial part of the proof is faithfulness.

We can always choose integers a_m , and we obtain a functor $\mathbf{Sp}_M(N) \rightarrow \mathbf{Sp}(N)$ to the spider category which sends $(m+k)^\pm \rightarrow (a_m+k)^\pm$ if $a_m+k \geq 0$ and otherwise sends this object to 0. Of course, this functor acts on diagrams by killing all diagrams containing heavy uprights with a label $m+k$ where $a_m+k < 0$, and otherwise turning heavy uprights with the label $m+k$ for $m \in M, k \in \mathbb{Z}$ into light uprights with label a_m+k .

Similarly, we have a functor from $U_{\mathbf{m}, \epsilon}$ to the mixed Schur algebra $\dot{U}_q(\mathfrak{gl}_\epsilon)_g$ as defined in [QS19, Prop. 7.1]. We thus have a commutative diagram:

$$\begin{array}{ccc} U_{\mathbf{m}, \epsilon} & \longrightarrow & \mathbf{Sp}_M(N) \\ \downarrow & & \downarrow \\ \dot{U}_q(\mathfrak{gl}_\epsilon)_g & \longrightarrow & \mathbf{Sp}(N) \end{array}$$

Consider the functor from top left to bottom right. No fixed one of these functors is faithful (as is clear from the definition), but ranging over all choices of a_m , they will be. By [QS19, Prop. 7.11], the bottom arrow is faithful, so we only need to consider the kernel of the left arrow. Thus, we need to show that for any non-zero element $x \in U_{\mathbf{r}' - \mathbf{r}} \otimes_{\mathbb{k}[\mathbf{K}^{\pm 1}]} \mathbb{k}_{\mathbf{r}}$, there exists a choice of a_m such that the image of x in $\dot{U}_q(\mathfrak{gl}_\epsilon)_g$ is non-zero.

By [QS19, Prop. 7.7], for any m , we have an action of $\dot{U}_q(\mathfrak{gl}_\epsilon)_g$ on a sum V of tensor product of wedge powers of $\mathbb{C}(q)^t$ and its dual, depending on the sign sequence ϵ , commuting with the action of $U_q(\mathfrak{sl}_t)$. This action is often referred to as “skew Howe duality.” Of course, we can understand the kernel of this action by considering the decomposition of V into irreducible representations of $\dot{U}_q(\mathfrak{gl}_n)$. An element x is in the kernel of this action if and only if it acts as zero on all irreducible representations that appear as summands of V . The irreducible representations of $\dot{U}_q(\mathfrak{gl}_n)$ that appear have a reasonable combinatorial description: we obtain exactly the irreps corresponding to highest weights λ satisfying $0 \leq \lambda_1 \leq \dots \leq \lambda_n \leq t$. The identity $1_{\mathbf{r}}$ acts by projection to a weight space for the action of $\dot{U}_q(\mathfrak{gl}_n)$, induced by the specialization $g = q^t, q^m = q^{a_m}$.

Thus, we have reduced to the question of showing that no nonzero element $x \in U_{\mathbf{r}'-\mathbf{r}} \otimes_{\mathbb{K}[\mathbf{K}^{\pm 1}]} \mathbb{K}_{\mathbf{r}}$ maps to zero under this specialization for all possible choices of t, a_m . So, for a given x , we need to find some t and a_m such that x acts non-trivially on some irreducible representation with highest weight λ satisfying $0 \leq \lambda_1 \leq \dots \leq \lambda_n \leq t$. We can consider the analogue of $U_{\mathbf{r}}$ in the subalgebras $U_q^\pm$ generated by E_i 's and F_i 's respectively. That is:

$$U_{(w_1, \dots, w_n)}^\pm = \{u \in U_q^\pm \mid K_i u K_i^{-1} = q^{w_i} u\}.$$

By the PBW theorem, we can assume without loss of generality that $x = x'x''$ for $x' \in U_{\mathbf{w}'}^+$ and $x'' \in U_{\mathbf{w}''}^-$ for some $\mathbf{w}', \mathbf{w}''$. We can choose a dominant weight λ' satisfying $0 \leq \lambda'_1 \leq \dots \leq \lambda'_n \leq t'$ for some t' where x' acts non-trivially on the highest weight vector of the corresponding highest-weight irrep $V_{\lambda'}$, and similarly a lowest weight λ'' satisfying $0 \leq \lambda''_n \leq \dots \leq \lambda''_1 \leq t''$ where x'' acts non-trivially on the lowest weight vector of the lowest weight irrep $\Lambda_{\lambda''}$. One can verify that this means that the action of $x = x'x''$ on their tensor product in $V_{\lambda'} \otimes V_{\lambda''}$ is non-zero. Choosing $t \geq t' + t''$ ensures all summands of this tensor product appear in V , so x acts non-trivially on V . $\square$

We can think about the resulting basis of Hom spaces more geometrically as follows. Fix $(m_1^{\epsilon_1}, \dots, m_n^{\epsilon_n})$. Let $\alpha_{ij} = (\delta_{jk} - \delta_{ik}) \in \mathbb{Z}^n$ be the usual root vector (so $\alpha_{12} = (-1, 1, 0, \dots)$, etc.). If we have non-negative integers $b_{ij} \in \mathbb{Z}_{\geq 0}$ for $i \neq j \in \{1, \dots, n\}$, which we collect into an $n \times n$ matrix B with zero diagonal, we let $w(B) = \sum_{i,j} b_{ij} \alpha_{ij}$.

Consider the ladder diagrams

$$(2.25) \quad c_{ij} = \begin{cases} \begin{array}{c} \begin{array}{ccccccc} & r_i - 1 & & & & & r_j + 1 \\ & \downarrow & \rightarrow & \downarrow & \rightarrow & \downarrow & \rightarrow \\ r_i & & & & & & r_j \\ & \downarrow & & \downarrow & & \downarrow & \\ & r_j + 1 & & & & & r_i - 1 \end{array} & i < j \\ - \begin{array}{c} \begin{array}{ccccccc} & r_j + 1 & & & & & r_i - 1 \\ & \downarrow & \leftarrow & \downarrow & \leftarrow & \downarrow & \leftarrow \\ r_j & & & & & & r_i \\ & \downarrow & & \downarrow & & \downarrow & \\ & r_j & & & & & r_i \end{array} & j < i \end{array} \end{cases}$$

For a matrix B as above, consider the morphism from $(r_1, \dots, r_n)$ to $(r_1 + w_1(B), \dots, r_n + w_n(B))$ given by

$$c(B) = \prod_{i=1}^n \prod_{j=1}^n c_{ij}^{b_{ij}}.$$

It follows from the PBW theorem and Lemma 2.11 that:

Corollary 2.12. *The space of morphisms*

$$\text{Hom}_{\text{Sp}_M(N)}((r_1, \dots, r_n), (s_1, \dots, s_n))$$

has a basis over $\mathbb{K}$ given by the diagrams $c(B)$ for B as above satisfying $w(B) = (r_1 - s_1, \dots, r_n - s_n)$.

2.3. Braids and cups. There is an important structure that arises from braiding heavy strands, though not a braided monoidal structure. Braids naturally act by permutations on n -tuples $\mathbf{m}, \epsilon$ by permutation, and in [QS19, Lem. 7.3], Queffelec and Sartori show that there is a braid equivalence $\mathbb{T}_\beta: \dot{U}_{\mathbf{m}, \epsilon} \rightarrow U_{\beta \mathbf{m}, \beta \epsilon}$ based on Lusztig's braid group action on the quantum group. We can thus transport this to an equivalence $\mathbb{T}_\beta: \text{Sp}_{\mathbf{m}, \epsilon}(N) \rightarrow \text{Sp}_{\beta \mathbf{m}, \beta \epsilon}(N)$. Geometrically, we can think of this as conjugating a ladder diagram by braiding the heavy uprights and then straightening the uprights. This will create crossings between rungs and uprights, which we can turn into ladders using the equations

(2.23–2.24), which reproduce the formulas of [QS19, (7.5)] up to sign. Note that if we interpret our algebra as factorization homology with marked points as in Remark 2.9, then this is just the action of the mapping class group of the marked surface on this factorization homology.

In fact, this action is most naturally thought of as an action of framed braids, where we have the option of adding any number of twists to each strand of the braid. This has the effect of just multiplying each diagram by a power of $g\nu_i^{-2}$, but is useful for bookkeeping purposes.

Similarly, while we cannot define a duality between heavy strands like we can between light strands with labels $k^\pm$, an important shadow of this structure exists in the form of modules over the category $\mathrm{Sp}_{\mathbf{m},\epsilon}(N)$.

Let us first describe this in the case of a single cup. Geometrically, C_m represents a heavy cup: two heavy strands joined at the bottom. The module structure encodes how light strands can interact with this cup—passing over the top produces the action of quantum group generators E and F .

We consider the category $U_{\mathbf{m},\epsilon}$ for $\mathbf{m} = (m, m)$ and $\epsilon = (+, -)$. In this case, there is a module C_m over $U_{\mathbf{m},\epsilon}$ sending the object (a, b) to a 1-dimensional vector space with basis v_p if $(a, b) = (m + p, -m - p)$ and to 0 otherwise, with the action:

$$(2.26) \quad Ev_p = \frac{q^{p+1}\nu - q^{-p-1}\nu^{-1}}{q - q^{-1}}v_{p+1} \quad Fv_p = \frac{gq^{-p+1}\nu^{-1} - g^{-1}q^{p-1}\nu}{q - q^{-1}}v_{p-1}$$

We visualize this more simply by representing v_p as a single cup joining uprights with labels $(p + m)^+$ and $(p + m)^-$. Thus the actions of E and F come from stringing a single light strand across the top of the cup either oriented left to right or the opposite.

$$(2.27) \quad \begin{array}{c} (m+p+1)^+ \quad (m+p+1)^- \\ \text{Diagram: A cup with a light strand crossing over from left to right.} \\ \text{Diagram: A cup with a light strand crossing over from right to left.} \end{array} = \frac{q^{p+1}\nu - q^{-p-1}\nu^{-1}}{q - q^{-1}} \begin{array}{c} (m+p+1)^+ \quad (m+p+1)^- \\ \text{Diagram: A cup with a light strand crossing over from left to right.} \\ \text{Diagram: A cup with a light strand crossing over from right to left.} \end{array}$$

$$(2.28) \quad \begin{array}{c} (m+p-1)^+ \quad (m+p-1)^- \\ \text{Diagram: A cup with a light strand crossing over from left to right.} \\ \text{Diagram: A cup with a light strand crossing over from right to left.} \end{array} = \frac{gq^{-p+1}\nu^{-1} - g^{-1}q^{p-1}\nu}{q - q^{-1}} \begin{array}{c} (m+p-1)^+ \quad (m+p-1)^- \\ \text{Diagram: A cup with a light strand crossing over from left to right.} \\ \text{Diagram: A cup with a light strand crossing over from right to left.} \end{array}$$

We can transport this to a module over $\mathrm{Sp}_{\mathbf{m},\epsilon}(N)$ by the equivalence of Lemma 2.11. It will be important for us to extend this to a module $\mathcal{C}\ell_m$ over $\mathrm{Spl}_{\mathbf{m},\epsilon}(N)$. Geometrically, this means that we allow ourselves to add light uprights with label 0 on either side of the diagram or in the middle of the cup, and allow rungs to join them from the heavy cup or to form light cups and caps. We impose the local relations of Definition 2.5 on these, and furthermore, allow light strands to isotope across the minimum of the heavy cup:

$$(2.29) \quad \begin{array}{c} (m+p+1)^+ \quad (m+p+1)^- \\ \text{Diagram: A cup with a light strand crossing over from left to right.} \\ \text{Diagram: A cup with a light strand crossing over from right to left.} \end{array} = \begin{array}{c} (m+p+1)^+ \quad (m+p+1)^- \\ \text{Diagram: A cup with a light strand crossing over from left to right.} \\ \text{Diagram: A cup with a light strand crossing over from right to left.} \end{array}$$

The relations Eqs. (2.27) and (2.28) are a consequence of these relations and (2.3).

We can also horizontally compose several of these cups to obtain a module $\mathcal{C}\ell_{m_1,\dots,m_k}$ over $\mathrm{Spl}_{\mathbf{m},\epsilon}(N)$ for $\mathbf{m} = (m_1, m_1, m_2, m_2, \dots, m_k, m_k)$ for some choice of $m_1, \dots, m_k$ and $\epsilon = (+, -, +, -, \dots, +, -)$. That is, we take the external tensor product of the modules C_{m_i} over the categories $\mathrm{Spl}_{m_i} = \mathrm{Spl}_{(m_i, m_i), (+, -)}(N)$ and then consider the induced module

$$C_{m_1,\dots,m_k} = \mathrm{Spl}_{\mathbf{m},\epsilon}(N) \otimes_{\mathrm{Spl}_{m_1} \otimes \dots \otimes \mathrm{Spl}_{m_k}} (C_{m_1} \otimes \dots \otimes C_{m_k}).$$

Geometrically, we can represent this as a sequence of k cups, each of which is labeled with $(m_i + p_i)^+$ and $(m_i + p_i)^-$ for different choices of $p_i \in \mathbb{Z}$ at the bottom of the cup, with additional light strands

joining the different cups.

$$(2.30) \quad \begin{array}{c} (m_1 + p_1)^+ \quad (m_1 + p_1)^- \\ \frown \\ (m_2 + p_2)^+ \quad (m_2 + p_2)^- \\ \frown \\ \dots \\ (m_k + p_k)^+ \quad (m_k + p_k)^- \\ \frown \end{array}$$

We obtain a module $C_{m_1, \dots, m_k}$ over $\mathrm{Sp}_{\mathbf{m}, \epsilon}(N)$ by only considering the diagrams with no light strands at the top; we need to define this module over $\mathrm{Spl}_{\mathbf{m}, \epsilon}(N)$ first so that we can use the relation (2.29) regardless of what the diagram on top of it looks like.

We can extend this construction by twisting the action on the module $C_{m_1, \dots, m_k}$ by the automorphism T_β attached to a framed braid β . This allows us to attach a left module ${}_T C$ to any framed tangle T with only minima; an analogous construction attaches a right module $C_{T'}$ to a tangle T' with only maxima.

Lemma 2.13. *The module attached to a framed tangle with only minima or only maxima is independent of the choice of braid β .*

Proof. To prove this, we only need to check that:

- (1) A full twist of the two strands at the top of a single cup sends $C_{m_1, \dots, m_k}$ to itself.
- (2) A crossing of two strands at the top of the j th cup over the strands at the top of the $(j+1)$ st cup sends $C_{m_1, \dots, m_k}$ to $C_{m_1, \dots, m_{j+1}, m_j, \dots, m_k}$.

Both of these follow from easy isotopy arguments.

2.4. Planar limit. We wish to consider how this construction behaves in the limit where we send $N \rightarrow \infty$ and $q \rightarrow 1$ so that $g = q^N$ remains fixed. Of course, we have already done half the work to take this limit—considering g as an independent parameter. Thus, it only remains to send $q \rightarrow 1$. In order to facilitate this, we will work over the base ring $\mathbb{k} = \mathbb{C}(q)[g^{\pm 1}, \nu^{\pm 1}]$; of course, this ring doesn't have a maximal ideal defined by $q = 1$, so we need to define a subcategory of $\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$ defined by the subring $\mathbb{k}' = \mathbb{C}[q]_{(q-1)}[g^{\pm 1}, \nu^{\pm 1}]$ where we replace $\mathbb{C}(q)$ by the subring with no poles at $q = 1$; note that this contains the q -binomial coefficients $\begin{bmatrix} k \\ l \end{bmatrix}_q$ for all $k, l \in \mathbb{Z}_{\geq 0}$, and the inverses of these coefficients if the corresponding binomial coefficient is non-zero. There are obvious difficulties in doing this, caused by the appearance of $q - q^{-1}$ in the denominators in numerous places. However, we can effectively just clear denominators everywhere and still obtain a well-defined limit.

Thus, in our diagrams, we will use a blue color and a dot-dash pattern on a rung in a ladder to indicate that the rung is multiplied by $q - q^{-1}$.

Convention: A blue, dot-dashed rung in a diagram indicates multiplication by $(q - q^{-1})$. This allows us to work over the subring $\mathbb{k}'$ where $q = 1$ is well-defined.

Of course, since this is just a scaling factor, only the number of blue rungs is significant, but since the power of $q - q^{-1}$ we need to multiply by is often proportional to the number of rungs in the ladder, this will prove to be a convenient shorthand. In particular, the equations (2.23–2.24) become the skein relation for blue rungs:

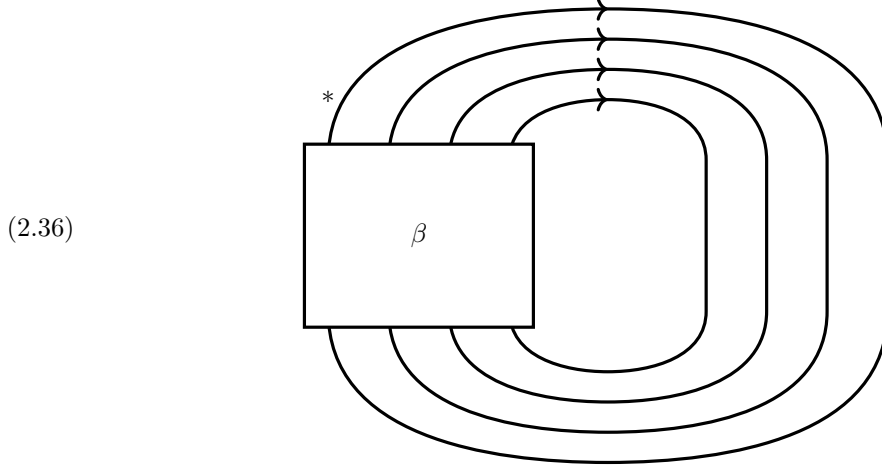
$$(2.31) \quad \begin{array}{c} 1 \quad m \\ \diagdown \quad | \\ \text{---} \quad \text{---} \\ \diagup \quad | \\ m \quad 1 \end{array} - \begin{array}{c} 1 \quad m \\ \diagdown \quad | \\ \text{---} \quad \text{---} \\ \diagup \quad | \\ m \quad 1 \end{array} = \begin{array}{c} 1 \quad m \\ \diagdown \quad | \\ \text{---} \quad \text{---} \\ \diagup \quad | \\ m \quad 1 \end{array}$$

(2.32)

Diagrammatic equation (2.32) showing the difference of two diagrams equals a third diagram. The first diagram has a vertical line with two vertices, each with a horizontal line (1) and a diagonal line (m). The second diagram has a vertical line with one vertex and a horizontal line (1), and a diagonal line (m) with one vertex. The third diagram has a vertical line with one vertex and a diagonal line (m) with one vertex. All lines are labeled with 1 and m.

The corresponding $\mathbb{C}[q, q^{-1}]$ lattice in $U_q(\mathfrak{gl}_k)$ is denoted by $U_q(\mathfrak{gl}_k)'$ and its construction is discussed in [Gav12, §7.3]. Recall that an **integral form** of a quantum group is a subalgebra defined over the Laurent polynomials $\mathbb{C}[q, q^{-1}]$ rather than the field $\mathbb{C}(q)$; this allows us to specialize q to non-zero complex numbers (such as $q = 1$) while maintaining algebraic structure. The limit as $q \rightarrow 1$ of this algebra is commutative and can be identified with the ring of functions on the dual group GL_k^* ; as an algebraic variety, this is isomorphic to $(\mathbb{C}^\times)^k \times \mathbb{C}^{k(k-1)}$. The coordinates for the first k factors are

label-shift operator Λ acts (shifting a light strand past the star requires applying Λ).



Having traced out a link L with a braid with heavy strands as above, we can then consider the formal span of all diagrams obtained by labeling the upward oriented portion of the heavy knot with elements of $M + \mathbb{Z}$, and adding in all possible configurations of light strands. We choose the convention that the elements of M are in bijection with the r components of the link L , and that upward portions of the heavy link are labeled by the corresponding element of M .

More formally, we can slice our braid closure in the middle, obtain left and right modules ${}_{T_1}C', C'_{T_2}$ over $\mathbf{Sp}'_{m,\epsilon}(N)$ corresponding to the top and bottom halves T_1, T_2 of the link.

Definition 2.15. We let $H_q(L) = {}_{T_1}C' \otimes_{\mathbf{Sp}'_{m,\epsilon}(N)} C'_{T_2}$.

This is independent of the choice of slicing, since any two choices differ by the action of a braid on T_1 and its inverse on T_2 . More generally, this is immediate from the isotopy invariance of skein evaluation. As discussed in the introduction, there is an element $[L] \in H_q(L)$, given by the heavy link with no light strands.

We can view this as a generalization of the computation of the HOMFLYPT polynomial as follows. For the usual HOMFLYPT polynomial with coloring $\bigwedge^{k_i} \mathbb{C}^N$ on each component, one specializes the formal parameters m_i to the integers k_i and evaluates in the spider category $\mathbf{Sp}(N)$. Our construction keeps m_i formal, which captures the relations between these polynomials for different colorings.

More precisely, we let L be the 1-dimensional representation of $U_q(\mathfrak{gl}_n \oplus \mathfrak{gl}_n)$ considered as a subalgebra of $U_q(\mathfrak{gl}_{2n})$ where E_i, F_i for $i \neq n$ act by 0 and K_i acts by 1 if $i < n$ and by g if $i > n$. We then let V be the parabolic Verma module induced from L with F_n acting by 0 (so this is a lowest weight module), considered as a left $U_q(\mathfrak{gl}_{2n})$ -module. We let V° be the right module obtained by twisting the action on V by the Cartan involution $E_i \mapsto F_i, F_i \mapsto E_i, K_i \mapsto K_i$.

Given $\mathbf{k} = (k_1, \dots, k_n) \in \mathbb{Z}_{\geq 0}$, there is a vector in this module $v_{\mathbf{k}}$ which corresponds to the fully nested cups T given by $E_n^{(k_n)} E_{n-1,n+2}^{(k_{n-1})} E_{n-2,n+3}^{(k_{n-2})} \cdots E_{1,n}^{(k_1)} v_0$.

We can compute the HOMFLYPT polynomial of a link L as written as the closure of a braid β on n strands as follows: the Lusztig braid action of β^{-1} induces a self-map $V \rightarrow V$ which intertwines the usual action of $U_q(\mathfrak{gl}_{2n})$ and an action twisted by this Lusztig braid action; let V^β be the parabolic Verma module with this twisted action. The tensor product $V^\circ \otimes V^\beta$ is 1-dimensional, spanned by $v_0 \otimes v_0$; the tensor product $v_{\mathbf{k}} \otimes v_{\mathbf{k}}$ is precisely the colored HOMFLYPT polynomial for the link L times $v_0 \otimes v_0$.

2.6. Label-shifting. The introduction of heavy objects introduces a new structure into this category: the ability to shift the label on heavy strands.

Definition 2.16. For any $m \in M$, the label-shift automorphism Λ_m of $\mathbf{Sp}_M(N)$ is an autoequivalence which:

- (1) Acts on objects by sending $(m+p)^\pm$ to $(m+p-1)^\pm$, and all other elements of $\Pi^\pm$ to themselves.
- (2) Acts on diagrams by shifting the label of each heavy upright as above.
- (3) Acts on the base ring $\mathbb{k}$ by

$$\Lambda_m(q) = q \quad \Lambda_m(g) = g \quad \Lambda_m(\nu') = q^{-\delta_{m,m'}} \nu.$$

Thus, label-shift automorphisms induce an action of the quantum torus $\mathcal{T}_r$ on the category $\mathbf{Sp}_M(N)$. We let $\widetilde{\mathbf{Sp}}_M(N)$ be the smash product category, where the morphisms are

$$\mathrm{Hom}_{\widetilde{\mathbf{Sp}}_M(N)}(\mathbf{r}, \mathbf{s}) \cong \bigoplus_{(a_1, \dots, a_r) \in \mathbb{Z}^r} \mathrm{Hom}_{\mathbf{Sp}_M(N)}(\mathbf{r}, \prod \Lambda_i^{a_i} \cdot \mathbf{s}).$$

We can extend all the modules appearing in the sections above to this category. In particular, $H_q(L)$ for a link L inherits a module structure over $\mathcal{T}_r$. If we consider the base change $H(L) = H_q(L) \otimes_{\mathbb{C}[q, q^{-1}]} \mathbb{C}$ where we set $q = 1$, this becomes a module structure over the Laurent polynomial ring $\mathbb{T}_r = \mathbb{C}[\nu_i^{\pm 1}, \Lambda_i^{\pm 1}, g^{\pm 1}]$. Over this algebra, we can interpret (2.34–2.35) as a presentation of the module C_m over the smash product:

$$((q - q^{-1})E - (\nu - \nu^{-1})\Lambda^{-1})v_0 = 0 \quad ((q - q^{-1})F - (gq\nu^{-1} - g^{-1}q^{-1}\nu)\Lambda)v_0 = 0$$

Recall that in the introduction, we defined an ideal

$$(2.37) \quad I_q(L) = \{x \in \mathcal{T}_r \mid x \cdot [L] = 0\}$$

This is the “ideal of relations between HOMFLYPT polynomials which can be derived from the MOY relations.”

In the introduction, we also introduced the space $\mathcal{F}_r$ of all maps from $\mathbb{Z}^r \rightarrow \mathbb{C}(q)[g^{\pm 1}]$, with its usual $\mathcal{T}_r$ -module structure. There is a $\mathcal{T}_r$ -module homomorphism $\mathrm{ev}: H(L) \rightarrow \mathcal{F}_r$ given by $\mathrm{ev}(D)(k_1, \dots, k_n)$ being the evaluation of D in the HOMFLYPT skein module, that is, as an endomorphism of the tensor identity in $\mathbf{Sp}(N)$, after the substitution $m_i \mapsto k_i$. Of course, under this specialization, we identify $\nu_i \mapsto q^{k_i}$.

Conjecture 2.17. *The map ev is injective and $I_q(L)$ is equal to the annihilator $I'_q(L)$ of $\mathrm{ev}([L])$.*

In [GLL18], Garoufalidis, Lê and Lauda show that $\mathrm{ev}([L])$ is a holonomic element of $\mathcal{F}_r$, that is, that the quotient $\mathcal{T}_r/I'_q(L)$ has Gelfand-Kirillov dimension r as a $\mathcal{T}_r$ -module.

3. COMPARISON WITH KNOT CONTACT HOMOLOGY

We want to compare the invariant $H(K)$ with knot contact homology.

3.1. Background on knot contact homology. Knot contact homology is a powerful knot invariant arising from symplectic geometry. Roughly speaking, it counts holomorphic disks in the cotangent bundle $T^*\mathbb{R}^3$ with boundary on the unit conormal bundle of the knot. For computational purposes, we use the combinatorial formulation of Ng [Ng14], which associates to a braid presentation of a knot a differential graded algebra (dga) whose homology is the knot contact homology.

The variable U used there is often denoted Q in more recent literature, but we will use U to more clearly match the notations of [Ng14]. This defines a dga $\mathcal{A}(K)$ over $\mathbb{C}[\mu^{\pm 1}, \lambda^{\pm 1}, U^{\pm 1}]$. The invariant of interest to us is the largest commutative quotient algebra of the degree 0 part of $\mathcal{A}(K)$. To spare ourselves unpleasantness with notation later, we will build in the change of variables that we will need later:

Definition 3.1. *Let $\mathrm{KCH}(K)$ be the largest commutative quotient algebra of the degree 0 part of $\mathcal{A}(K)$ with scalars extended from $\mathbb{C}[\mu^{\pm 1}, \lambda^{\pm 1}, U^{\pm 1}]$ to $\mathbb{T}_1 = \mathbb{C}[\nu^{\pm 1}, \Lambda^{\pm 1}, g^{\pm 1}]$ via the injective map*

$$(3.1) \quad \mu \mapsto \nu^{-2} \quad U \mapsto g^{-2} \quad \lambda \mapsto -g^{-1}\Lambda^{-1}.$$

This is the coordinate ring of the full augmentation variety of the knot K in the sense of Diogo and Ekholm [DE25]; it could thus also be called the full augmentation algebra. The full augmentation variety maps to the usual augmentation variety by forgetting the variables a_{ij} , which correspond to augmentations of Reeb chords between different strands of the braid. Thus, the augmentation algebra is the subalgebra of $\mathrm{KCH}(K)$ generated by $\nu^{\pm 1}, \Lambda^{\pm 1}, g^{\pm 1}$.

The algebra $\mathcal{A}(K)$ is concentrated in degrees ≥ 0 , with the convention that the degree of the differential is -1 (contrast to the majority of the literature on dg-algebras). Thus, $\mathrm{KCH}(K)$ is generated by Ng’s generators of degree 0 $a_{ij}, \mu^{\pm 1}, \lambda^{\pm 1}, U^{\pm 1}$ subject to the relations that the resulting algebra is commutative and the differential of any element of degree 1 is 0. Let $\mathcal{F}$ be the free commutative algebra over $\mathbb{C}[\mu^{\pm 1}, \lambda^{\pm 1}, U^{\pm 1}]$ generated by a_{ij} with $i \neq j$.

The algebra $\mathcal{F}$ inherits a braid group action from [Ng14, Def. 3.3], where the positive braid generator σ_k acts by:

$$(3.2) \quad \phi_{\sigma_k} : \begin{cases} a_{ij} \mapsto a_{ij}, & i, j \neq k, k+1 \\ a_{k+1,i} \mapsto a_{ki}, & i \neq k, k+1 \\ a_{i,k+1} \mapsto a_{ik}, & i \neq k, k+1 \\ a_{k,k+1} \mapsto -a_{k+1,k} \\ a_{k+1,k} \mapsto -a_{k,k+1} \\ a_{ki} \mapsto a_{k+1,i} - a_{k+1,k}a_{ki}, & i \neq k, k+1 \\ a_{ik} \mapsto a_{i,k+1} - a_{ik}a_{k,k+1}, & i \neq k, k+1. \end{cases}$$

We now introduce matrices that organize the generators a_{ij} in a way that makes the connection to our skein-theoretic constructions transparent. Define $n \times n$ matrices $\mathbf{A}, \hat{\mathbf{A}}$ in $\mathcal{F}$ by:

$$\mathbf{A}_{ij} = \begin{cases} a_{ij} & i < j \\ -\mu a_{ij} & i > j \\ 1 - \mu & i = j \end{cases} \quad \hat{\mathbf{A}}_{ij} = \begin{cases} U a_{ij} & i < j \\ -\mu a_{ij} & i > j \\ U - \mu & i = j \end{cases}$$

These matrices encode the augmentation data: the diagonal entries record the meridian variable μ , while off-diagonal entries a_{ij} correspond to “paths” between strands i and j that contribute to the augmentation variety. The asymmetry between $i < j$ and $i > j$ reflects the orientation of the braid.

The braid group action above can be captured by degree n representations $B_n \rightarrow GL_n(\mathcal{F})$

$$\beta \mapsto \Phi_\beta^L \quad \beta \mapsto (\Phi_\beta^R)^{-1}$$

defined by the formulas below:

$$\Phi_{\sigma_i}^L = \left[\begin{array}{c|cc|c} I_{i-1} & 0 & 0 & 0 \\ \hline 0 & -a_{i+1,i} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & I_{n-i-1} \end{array} \right] \quad \Phi_{\sigma_i}^R = \left[\begin{array}{c|cc|c} I_{i-1} & 0 & 0 & 0 \\ \hline 0 & -a_{i,i+1} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & I_{n-i-1} \end{array} \right]$$

which satisfies $\phi_\beta(\mathbf{A}) = \Phi_\beta^L \mathbf{A} \Phi_\beta^R$. Consider the diagonal matrix $\mathbf{\Lambda} = \text{diag}(\lambda \mu^{\text{wr}(\beta)} U^{(n-\text{wr}(\beta)-1)/2}, 1, \dots, 1)$.

Lemma 3.2. *Under the map $\mathcal{F} \rightarrow \mathbf{Sp}_{\mathbf{m},\epsilon}(N)$ sending $a_{ij} \mapsto \gamma_{ij}$, this braid group action is intertwined with the braid group action of Section 2.3.*

Before the proof of this result, let us record a set of relations that show how we can change framing, which have already appeared in [GKS25, (E.21-23)]:

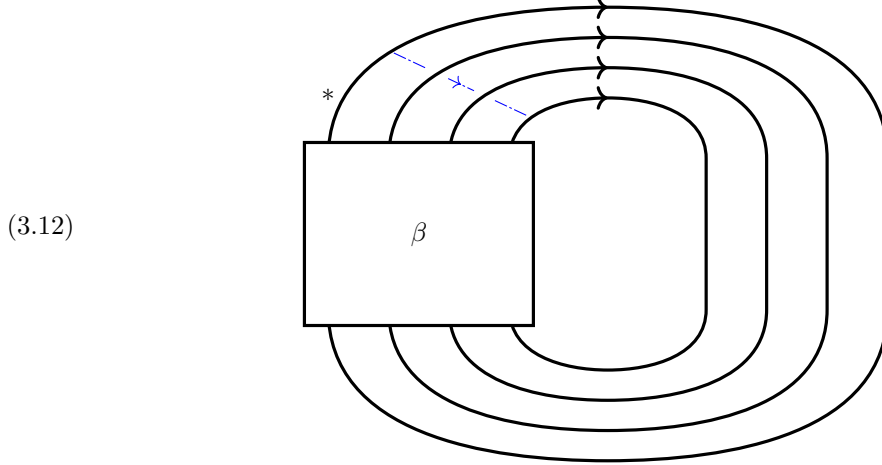
$$(3.3) \quad \text{Diagram of a loop with a crossing} = -g \text{Diagram of a straight strand}$$

$$(3.4) \quad \text{Diagram of a loop with a crossing and a strand} = -g^{-1}\nu \text{Diagram of a straight strand} \quad \text{Diagram of a loop with a crossing and a strand} = -g\nu^{-1} \text{Diagram of a straight strand}$$

Proof. This is a straightforward computation. The case $i, j \neq k, k+1$ is simple isotopy ignoring the crossing strands. The case of $a_{k+1,i}$ and $a_{i,k+1}$ for $i \neq k, k+1$ is an isotopy along the $(k+1)$ th strand as shown below:

$$(3.5) \quad \text{Diagram of a crossing with strands } m_{i+1}, m_i, m_i, m_{i+1} = \text{Diagram of a crossing with strands } m_{i+1}, m_i, m_i, m_{i+1} \quad \text{Diagram of a crossing with strands } m_{i+1}, m_i, m_i, m_{i+1} = \text{Diagram of a crossing with strands } m_{i+1}, m_i, m_i, m_{i+1}$$

In the diagram below, we show the image of a_{14} for a 4-strand braid:



Remark 3.6. This is the first point where the fact that K is a knot is used: if K is a link and we draw a light strand connecting two different components of K , then there is no consistent way of labeling the heavy strands, and thus this does not give a well-defined element of $H(K)$. We discuss in Section 3.3 how to modify this construction to work for links.

Note that in our notation, $\mathbf{\Lambda} = \text{diag}(\Lambda^{-1}\nu^{-2\text{wr}(\beta)}(-g)^{\text{wr}(\beta)-n}, 1, \dots, 1)$. Since we assume that K is a knot, this means that the permutation of the strands induced by the braid β must be a single cycle of length n . Thus, for each i , there is a unique integer $d(i) \in \{0, \dots, n-1\}$ such that $1 = \beta^{d(i)}(i)$. Note that

$$(3.13) \quad d(\beta(i)) = \begin{cases} d(i) - 1 & i \neq 1 \\ n - 1 & i = 1 \end{cases}$$

Let $k(i, j) = d(j) - d(i)$. Let us rewrite the definitions above in terms that will be more convenient for us. Let $D = \text{diag}((-g)^{d(1)}, \dots, (-g)^{d(n)})$. Consider the automorphism Ψ of $\mathcal{F}$ that sends $\mathbf{A} \rightarrow D\mathbf{A}D^{-1}$, that is $a_{ij} \mapsto (-g)^{-k(i,j)}a_{ij}$. Let $D_\beta = \text{diag}((-g)^{d(\beta(1))}, \dots, (-g)^{d(\beta(n))})$. Note that

$$-g\mathbf{\Lambda}D_\beta D^{-1} = \mathbf{\Lambda}' := \text{diag}(\Lambda^{-1}\nu^{-2\text{wr}(\beta)}(-g)^{\text{wr}(\beta)}, 1, \dots, 1)$$

We wish to consider how this automorphism affects the matrices $\Phi_\beta^L, \Phi_\beta^R$.

Lemma 3.7.

$$\Psi(\Phi_\beta^L) = D_\beta \Phi_\beta^L D^{-1} \quad \Psi(\Phi_\beta^R) = D \Phi_\beta^R D_\beta^{-1}$$

Proof. One can easily check by induction that the ij -entry of Φ_β^L is a sum of monomials $a_{\beta(i), j_1} a_{j_1, j_2} \cdots a_{j_\ell, j}$ for different sequences $j_1, \dots, j_\ell$. The automorphism Ψ acts on these by multiplication by $(-g)^{-k(\beta(i), j)}$. The result follows for Φ_β^L . The argument for Φ_β^R is identical with indices reversed. $\square$

Thus, when we apply Ψ to the equations (3.8–3.10), we obtain the modified relations:

$$(3.14) \quad \mathbf{A} = \mathbf{\Lambda}' \cdot \phi_\beta(\mathbf{A}) \cdot (\mathbf{\Lambda}')^{-1}$$

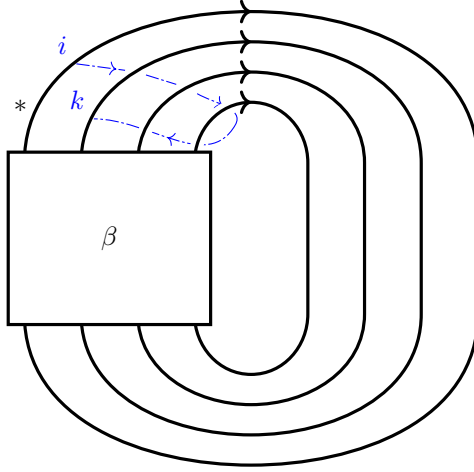
$$(3.15) \quad -g\hat{\mathbf{A}} = \mathbf{\Lambda}' \cdot \Phi_\beta^L \cdot \mathbf{A}$$

$$(3.16) \quad \mathbf{A} = -g\hat{\mathbf{A}} \cdot \Phi_\beta^R \cdot (\mathbf{\Lambda}')^{-1}$$

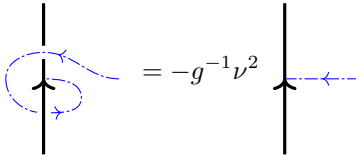
We can rewrite the matrices $\mathbf{A}$ and $\hat{\mathbf{A}}$ identifying the formal symbols a_{ij} with diagrams γ_{ij} defined in (2.33). It will be more convenient for our proof to consider $-g\hat{\mathbf{A}}$, so we write out the entries in our conventions:

$$\mathbf{A}_{ij} = \begin{cases} a_{ij} & i < j \\ -\nu^{-2}a_{ij} & i > j \\ 1 - \nu^{-2} & i = j \end{cases} \quad -g\hat{\mathbf{A}}_{ij} = \begin{cases} -g^{-1}a_{ij} & i < j \\ g\nu^{-2}a_{ij} & i > j \\ g\nu^{-2} - g^{-1} & i = j \end{cases}$$

Now, we slide the second half of the light strand upward through the braid to obtain a sum of diagrams α_{ik} for different values of k . By [Ng14, Def. 3.9], this sum is exactly described by the matrix Φ_β^R , as the sum

$$(3.23) \quad \mathbf{A}_{ij} = \sum_{k=1}^n (\Phi_\beta^R)_{jk}$$


This result is very close to $-g\hat{\mathbf{A}} \cdot \Phi_\beta^R$, but if $k = 1$, we have to drag past the star and $-\text{wr}(\beta)$ twists in the heavy strand. Together this is the same as multiplying by $(\Lambda')^{-1}$ by (3.11) and the relation:

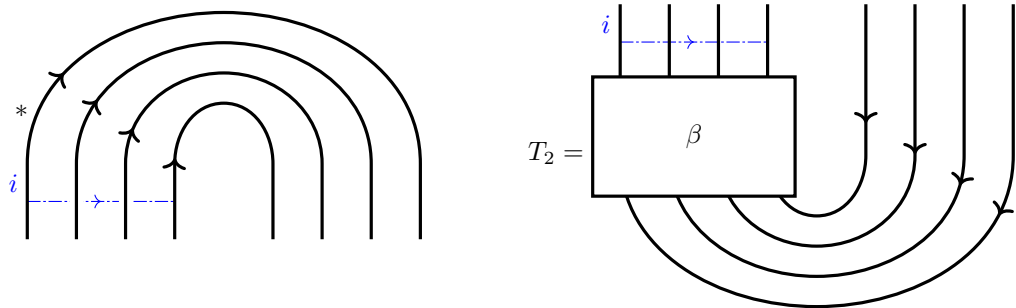
$$(3.24) \quad \text{Diagram} = -g^{-1}\nu^2 \text{Diagram}$$


This proves (3.16).

A similar argument applies to (3.15), sliding the first half of the light strand rather than the second half. Since we are sliding the first half through the braid, we get the action of Φ_β^L on the left, but otherwise the argument is the same. The equation (3.14) follows from substituting (3.15) into (3.16). $\square$

Lemma 3.9. *The map Z is an isomorphism.*

Proof. To prove this, we need only construct an inverse map $Y: \mathbf{H}(K) \rightarrow \mathbf{KCH}(K)$. By the universal property of tensor products, we have a $\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$ -bilinear map $\tilde{Y}: {}_{T_1}C' \times C'_{T_2} \rightarrow \mathbf{KCH}(K)$ where T_1 is a tangle given by n nested caps and T_2 by n nested cups, with their left ends braided by β . The tangles are shown below with a representative light strand.

$$(3.25) \quad T_1 = \text{Diagram} \quad T_2 = \text{Diagram}$$


Since ${}_{T_1}C'$ is a limit of an induction, we can give it a basis by

- (1) Multiplying on the leftmost n strands by a monomial in a_{ij} for $i \neq j$, as shown in the picture above.
- (2) Fixing the labels on the left ends of the strands to be $m + p_1, \dots, m + p_n$; the labels on the right ends are determined by the pattern of the light strands.

Similarly, we can give a basis to N_{T_2} by similar multiplication and fixing labels. Let us emphasize that this is done above the braid. For $a \in \mathcal{F}$ and $\mathbf{p}$, let $_{\mathbf{p},a}v$ be the corresponding vector in $_{T_1}N$ and $v_{\mathbf{p},a} \in N_{T_2}$.

Thus, we can define the map $\tilde{Y}$ by taking the unique $\mathbb{T}_1$ -linear map such that $\tilde{Y}(_{\mathbf{p},a}v, v_{\mathbf{p}',a'}) = 0$ if the labels on the bottom of $_{\mathbf{p},a}v$ don't match those on top of $v_{\mathbf{p}',a'}$, and $\tilde{Y}(_{\mathbf{p},a}v, v_{\mathbf{p}',a'})$ is the image of $\Lambda^{-p_1}aa'$ in $\text{KCH}(K)$ otherwise.

Now we only need to confirm $\mathbf{Sp}_{\mathbf{m},\epsilon}(N)$ -bilinearity. That is, we have to prove the compatibility

$$\tilde{Y}(_{\mathbf{p},a}v \cdot \gamma_{ij}, v_{\mathbf{p}',a'}) = \tilde{Y}(_{\mathbf{p},a}v, \gamma_{ij} \cdot v_{\mathbf{p}',a'})$$

with multiplication by γ_{ij} for $i, j \in \{1, \dots, 2n\}$.

- (1) If $i, j \leq n$, then this is manifest from the definition.
- (2) If $i \leq n < j$, then this follows from (3.10). The diagram γ_{ij} is an intermediate step in the proof of this equality; the first half uses equalities in $_{T_1}N$ and the second half equalities in N_{T_2} .
- (3) If $j \leq n < i$, then this follows by the same argument as for (3.9).
- (4) If $n < i, j$, then this follows by the same argument from (3.8). $\square$

This completes the proof of Theorem 3.4.

3.3. Links. If, instead of a knot, we consider a link $L = L_1 \cup \dots \cup L_r$ with multiple components, then the results of the paper continue with minor changes. Let us list here the changes that need to be made to carry this out in the case of links to compare with the knot contact homology defined for links in [Ng14, Appendix].

- (1) The definition of $\mathbf{H}(L)$ remains unchanged, but we can now independently choose the label m_i on the different components, which we number in order from left to right in terms of where we encounter their leftmost strand. Thus, the vector space $\mathbf{H}(L)$ is now a module over the ring $\mathbb{T}_r$.
- (2) We now add the correct number of twists to produce the zero-framing of each component on the leftmost strand lying in that component, just above the braid as before.
- (3) We now define the integer $d(i)$ to be the smallest non-negative integer such that $\beta^{d(i)}(i)$ is the leftmost strand of its component, and we define the automorphism Ψ by

$$\Psi(a_{ij}) = (-g)^{d(j)-d(i)} \frac{\mu_i}{\mu_j} a_{ij}.$$

- (4) With this modification, the relations (3.8–3.10) hold for the diagonal matrix $\mathbf{\Lambda}'$ whose entries are 1 except for those corresponding to the leftmost strand of each component, which are $\Lambda_i^{-1} \nu^{2 \text{wr}(L_i)} (-g)^{\text{wr}(L_i)}$, where $\text{wr}(L_i)$ is the writhe of the diagram of L_i obtained by forgetting all other components.

- (5) The map $Y: \mathbf{H}(L) \rightarrow \text{KCH}(L)$ is well-defined by the same arguments as in Lemma 3.9.

Unfortunately, the map Y is not manifestly surjective. For each component L_i , we let k_i be the number of strands beginning on the i th component minus the number of strands ending on it. The vector space $\text{KCH}(L)$ is graded by $\mathbb{Z}^r$, where the grading of a diagram is $(k_1, \dots, k_r)$. Since $\sum k_i = 0$, this grading is trivial if $r = 1$, but for $r > 1$, it is not clear to us whether it is trivial. Thus, applying the same arguments again shows that:

Theorem 3.10. *The map $Y: \mathbf{H}(L) \rightarrow \text{KCH}(L)$ is an isomorphism to the degree 0 subspace of $\text{KCH}(L)$.*

Remark 3.11. It has been noted previously that the spectrum of $\text{KCH}(L)$ can have dimension which is too large for a link; for example, [Gao21, §4.4] describes a 4-dimensional component for the 3-component unlink. It would be interesting to consider whether this degree 0 subspace points the way to a better definition for links which has the expected dimension.

3.4. Comparison with the augmentation ideal. Theorem 3.10 shows that the annihilator in the ring $\mathbb{T}_r$ of the class $[L]$ in $\mathbf{H}(L)$ is the same as the augmentation ideal Au_L . However, this is not obviously the same as the ideal $I(L)$ obtained by base-changing $I_q(L)$; we cannot rule out the possibility that it is larger. It is helpful to understand this by considering the exact sequence of $\mathbb{C}[q^{\pm 1}]$ -modules $0 \rightarrow \mathcal{T}_r/I_q(L) \rightarrow \mathbf{H}_q(L) \rightarrow \mathbf{H}_q(L)/\mathcal{T}_r \cdot [L] \rightarrow 0$. The universal coefficient theorem gives us a six-term

exact sequence:

$$0 \rightarrow \operatorname{Tor}_{q-1}(\mathcal{T}_r/I_q(L)) \rightarrow \operatorname{Tor}_{q-1}(\mathbf{H}_q(L)) \rightarrow \operatorname{Tor}_{q-1}(\mathbf{H}_q(L)/\mathcal{T}_r \cdot [L]) \xrightarrow{\partial} \mathcal{T}_r/I(L) \rightarrow \mathbf{H}(L) \rightarrow \mathbf{H}(L)/\mathcal{T}_r \cdot [L] \rightarrow 0$$

where $\operatorname{Tor}_{q-1}(M) = \{m \in M \mid (q-1)m = 0\}$ for any $\mathbb{C}[q^{\pm 1}]$ -module M . The quotient $\mathbf{Au}_L/I(L)$ is the image of the connecting homomorphism ∂ . We expect that this connecting homomorphism ∂ is 0 for all links L . Combined with Conjecture 2.17, this would imply that:

Conjecture 3.12. *We have equalities $I_q(L) = I'_q(L)$ and $I(L) = \mathbf{Au}_L$ for all links L . That is, the augmentation ideal $\mathbf{Au}_L$ is the classical limit of the ideal defining the recursion relations on antisymmetric HOMFLYPT polynomials.*

In particular, the vanishing set of the augmentation ideal $\mathbf{Au}_L$ is Lagrangian in $(\mathbb{C}^\times)^{2r}$.

In both cases, our ability to prove this conjecture is limited by our poor understanding of the structure of the module $\mathbf{H}_q(L)$. If we knew, for example, that this module was free as a $\mathbb{C}[q^{\pm 1}, \nu_i^{\pm 1}]$ -module of known rank, then our task would be considerably simplified.

INDEX OF NOTATION

This index of notation gives a brief description of the main notation used in the paper, together with the section and page where the notation is defined.

§	Symbol	Description	Page
2.1	g	The quantity q^N where we think of N as a variable.	3
	$\mathbf{Sp}(N)$	The HOMFLYPT spider category	3
2.2	M	The index set of we'll use for labels on heavy uprights.	7
	$\mathbb{k}$	The base ring $\mathbb{C}(q)[g^{\pm 1}, q^{\pm m}]$.	7
	ν, ν_i	The formal exponential q^m, q^{m_i} .	7
	Π	The combined index sets for light and heavy uprights $\Pi = \mathbb{Z}_{\geq 0} \cup M + \mathbb{Z}$.	7
	$\mathbf{Sp}_M(N)$	The spider category incorporating heavy uprights labeled with elements of Π .	7
	$\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$	The full subcategory of $\mathbf{Sp}_M(N)$ consisting of the objects $((m_1 + p_1)^{\epsilon_1}, \dots, (m_n + p_n)^{\epsilon_n})$.	8
	$\mathbf{Spl}_{\mathbf{m}, \epsilon}(N)$	The full subcategory of $\mathbf{Sp}_M(N)$ consisting of the objects where the labels on heavy uprights are $((m_1 + p_1)^{\epsilon_1}, \dots, (m_n + p_n)^{\epsilon_n})$, but additional light uprights are allowed.	8
	$\dot{U}_{\mathbf{m}, \epsilon}$	The extension of the idempotent quantum group to include the objects $((m_1 + p_1)^{\epsilon_1}, \dots, (m_n + p_n)^{\epsilon_n})$.	10
2.3	T_β	The equivalence $T_\beta: \mathbf{Sp}_{\mathbf{m}, \epsilon}(N) \rightarrow \mathbf{Sp}_{\beta\mathbf{m}, \beta\epsilon}(N)$ induced by a braid β .	11
	C_m	The module over $\mathbf{Sp}_{(m, m), (+, -)}(N)$ corresponding to a single cup.	12
	$T C, C_{T'}$	The $\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$ module corresponding to a tangle T with only minima or a tangle T' with only maxima.	13
2.4	$\mathbb{k}'$	The subring $\mathbb{C}[q]_{(q-1)}[g^{\pm 1}, \nu^{\pm 1}] \subset \mathbb{k}$ where we only use rational functions regular at $q = 1$.	13
	$\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$	The $\mathbb{k}'$ -lattice in $\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$ generated by the diagrams γ_{ij} .	14
2.5	$C'_{m_1, \dots, m_k}$	The $\mathbf{Sp}_{\mathbf{m}, \epsilon}(N)$ obtained by specializing $C_{m_1, \dots, m_k}$.	14
	$\operatorname{wr}(\beta)$	The writhe of the braid β .	14
	$\mathbf{H}_q(L)$	The link invariant $\mathbf{H}_q(L) = T_1 C' \otimes_{\mathbf{Sp}'_{\mathbf{m}, \epsilon}(N)} C'_{T_2}$.	15
	$[L]$	The element of $\mathbf{H}_q(L)$ given by the heavy link with no light strands.	15
2.6	$\mathbf{H}(L)$	The base change of $\mathbf{H}_q(L)$ where we set $q = 1$.	16
	$\mathcal{T}_r$	The Laurent polynomial ring $\mathbb{C}[\nu_i^{\pm 1}, \Lambda_i^{\pm 1}, g^{\pm 1}]$	16
	$I_q(L)$	The ideal in $\mathcal{T}_r$ annihilating $[L]$.	16

§	Symbol	Description	Page
3.1	$\text{KCH}(K)$	The largest commutative quotient algebra of the degree 0 part of knot contact homology dga $\mathcal{A}(K)$.	16
	μ, U, λ	The usual variables for knot contact homology, related to ours by $\mu = \nu^{-2}, U = g^{-2}, \lambda = -g^{-1}\Lambda^{-1}$.	16
	$\mathcal{F}$	The free commutative algebra over $\mathbb{C}[\mu^{\pm 1}, \lambda^{\pm 1}, U^{\pm 1}]$ generated by a_{ij} with $i \neq j$.	16
	ϕ_β	The action of a braid β on the algebra $\mathcal{F}$.	17
	$\mathbf{A}, \hat{\mathbf{A}}, \mathbf{\Lambda}$	Matrices in $\mathcal{F}$ that appear in the relations of $\text{KCH}(K)$.	17
	$\Phi_{\sigma_i}^L, \Phi_{\sigma_i}^R$	Representations of the braid group which describe the action on the matrix $\mathbf{A}$.	17
3.2	$d(i), k(i, j), D, D_\beta$	Statistics and a diagonal matrix that measure how many iterations of β send i to 1.	19

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